TITLE PLACEHOLDER

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Abstract

ABSTRACT PLACEHOLDER

Introduction

Results

Discussion

Methods

Our experimental protocol was approved by the Committee for the Protection of Human Subjects at Dartmouth College. We obtained written informed consent from all participants prior to the start of both experiment sessions.

Participants

We enrolled a total of 70 Dartmouth undergraduate students in our study. Participants received optional course credit for enrolling and were pre-screened to ensure they had not previously seen any episode of either TV show viewed during the experiment. No statistical method was used to predetermine sample size. Prior to data processing and analysis, we excluded a total of 17 participants from our dataset: 4 encountered technical issues with the experiment, 2 self-reported difficulty with the task due to illness, 4 spoke for less than 10 minutes during the immediate recall phase, and 7 did not return for the second experiment session. The data we analyzed comprised the remaining 53 participants.

At the start of the experiment, we asked each participant to complete a demographic survey that included questions about their age, sex, race, and ethnicity. Participants' ages ranged from 18 to 22 years (mean: 18.91 years; standard deviation: 1.01 years). A total of 41 participants reported their sex as female and 12 participants reported their sex as male. Participants reported their races as White (34 participants), Asian (16 participants), Black or African American (3 participants), American Indian or Alaska Native (1 participant), and other (1 participant). (Note that some participants selected multiple racial categories.) A total of 52 participants reported their ethnicity as "Not Hispanic or Latino" and 1 participant reported their ethnicity as "Hispanic or Latino."

We also asked participants their native spoken language; whether they had any hearing, speech, or color vision impairments; and about any current medications or recent injuries that could impact attention or memory. All participants reported that their native spoken language was English, that they had no hearing or speech impairments, and that they had normal color vision. A total of 52 participants reported no attention- or memory-relevant medications or injuries. One participant reported sustaining a concussion more than 3 years prior to the experiment and was not excluded from our dataset.

Additionally, before each of the two experiment sessions, we asked participants about their sleep, coffee consumption, and level of alertness. Participants reported having slept between 5 and 11 hours the night prior to session 1 (mean: 7.37 hours; standard deviation: 1.45 hours) and between 5 and 11 hours the night prior to session 2 (mean: 7.39 hours; standard deviation: 1.23 hours). They reported having consumed between 0 and 3 cups of coffee on the day of session 1 (mean: 0.43 cups; standard deviation: 0.72 cups) and between 0 and 3 cups of coffee on the day of session 2 (mean: 0.48 cups; standard deviation: 0.68 cups). Participants reported their current levels of alertness on a 5-point Likert scale. At the start of session 1, 1 participant reported feeling "Very sluggish," 14 reported feeling "A little sluggish," 18 reported feeling "Neutral," 15 reported feeling "A little alert," and 5 reported feeling "Very alert." At the start of session 2, 3 participants reported feeling "Very sluggish," 9 reported feeling "A little sluggish," 23 reported feeling "Neutral," 17 reported feeling "A little alert," and 3 reported feeling "Very alert."

@jeremy: *do we want to report class years and majors? And/or do we want to report post-experiment survey responses (how engaging did you find the episode, how easy/difficult was it to follow, how well do you think you recalled it, how well do you think you learned characters' names)?*

Experiment

The experiment took place over two sessions lasting approximately one hour each, separated by 6–8 days (mean: 6.76 days; standard deviation: 0.73 days). Prior to the start of the experiment, participants were assigned to one of two groups (*A* or *B*), alternating based on enrollment order. During session 1, all participants first viewed the pilot episode of the FX series *Atlanta* ("The Big Bang"; duration: 24 min 15 s). Immediately after viewing, participants were given up to one hour to recount what happened in the episode to the best of their abilities. Following Chen et al. (2017), participants were instructed to describe the episode in as much detail as possible and to try to speak for a minimum of 10 minutes. They were told to try to recount the episode's events in their original order, but that they were free to go back and describe earlier events should they realize they had missed something. After the immediate recall phase, participants were then given up to 5 minutes to predict what they thought would happen in the show's next episode.

Upon returning for session 2, all participants were again given up to one hour to recount the *Atlanta* episode they had viewed the week prior. Following the delayed recall phase, participants in Group *A* then viewed the second episode of *Atlanta* ("Streets on Lock"; duration: 21 min 57 s) while participants in Group *B* instead viewed the pilot episode of the FOX series *Arrested Development* ("Pilot"; duration: 20 min 37 s). Both groups were then given up to one hour to recount the just-watched episode in the same manner and given the same instructions as in session 1. We implemented the experiment using the psiTurk platform (Gureckis et al., 2016) and jsPsych library (de Leeuw, 2015), with a custom jsPsych plugin we developed in prior work for recording verbal recall audio (Ziman et al., 2018). Here we analyze participants' immediate and delayed recalls of the *Atlanta* episode viewed in session 1.

Stimulus annotation and recall transcription

To capture the evolving semantic content of the *Atlanta* episode, we created detailed text annotations of the on-screen content during each individual camera shot throughout the episode (mean annotation duration: 3.05 s; standard deviation: 1.80 s). Specifically, we used ANVIL (Kipp, 2001) to record, for each camera shot, a set of semantically relevant features. These included a brief description of physical occurrences and characters' explicit actions, a brief description of characters' internal states (e.g., emotions, motivations) and other implicit content, the names of all characters appearing on screen, a transcription of any dialogue, the name(s) of the character(s) speaking, the shot's setting, whether the shot took place indoors or outdoors, and whether or not there was music playing during the shot. This feature set was adapted from a similar annotation corpus created by Chen et al. (2017), which we used in developing our trajectory-analysis framework in prior work (Heusser et al., 2021).

We then transcribed each participant's immediate and delayed verbal recall of the *Atlanta* episode. Our transcription process consisted of three steps: an initial manual transcription, a two-pass prose-formatting conversion with a large language model, and a final manual cleanup and correction step. First, we manually transcribed each recall audio file collected during the experiment. These initial transcripts reflected participants' spoken recalls verbatim, without any punctuation or additional formatting, and including disfluencies, stuttered repetitions, and other artifacts common to unrehearsed speech. However, since transformer models (such as we used to obtain text embeddings of participants' recalls) are trained almost exclusively on written text, structural conventions like sentence breaks, punctuation, and capitalization provide an important signal these models leverage to contextual other tokens (e.g., words) and construct a representation of the text's overall meaning. Meanwhile, speech-native phenomena such as stuttered repetitions and filler words ("like", "um", etc.) that are comparatively rare in written text contribute out-of-distribution noise that can be expected to degrade embedding fidelity, and prior work has shown that automated removal of these phenomena improves performance on downstream tasks that assume clean, written-style input (Chen et al., 2022; Wang et al., 2010). We therefore used a large language model (Qwen3.5 122b; Qwen Team, 2026) to convert participants' recall transcripts to a structure and style more typical of written text. We used the `langchain_dartmouth` Python package (Stone et al., 2025) to implement this conversion as two sequential passes over each transcript. In the first pass, we prompted the model to split the transcript into sentences at natural breaking points, add punctuation including quotes around reported speech, remove clear non-word disfluencies ("um," "uh,", "hmm," etc.), collapse stuttered verbatim repetitions, and mark self-corrections and restarts with em-dashes. In the second pass, we prompted the model to remove all instances of "like" used as a filler word. We removed filler "like" instances in a separate pass because of their high frequency of occurrence in the transcripts and the additional effort to required to distinguish them from meaningful uses of "like." The full texts of our prompts for both passes can be found in *Supplementary methods*. Finally, we manually reviewed all LLM-formatted transcripts to identify and correct any errors, ensure sentence breaks were appropriate and consistent across participants, confirm that no content was improperly added to or removed from the transcripts, and resolve any ambiguities based on the prosody of participants' original recall audio.

Analysis

Statistics

Wilcoxon tests were one-sided because our alternative hypothesis was directional (delayed matches own immediate *better than* the episode, others' immediates, etc.).

Constructing episode and recall trajectories

We adapted a method we developed in prior work (Heusser et al., 2021) to transform the *Atlanta* episode and participant's recalls of it into trajectories through a common representation space using a text embedding model. Briefly, text embedding models take in natural language text of an arbitrary length and output a fixed-length vector representation that captures the text's semantic meaning. Intuitively, an output vector of length n can be viewed as a coordinate in an n-dimensional text embedding space. While specific model architectures and training paradigms vary, modern text embedding models are broadly trained to output similar coordinate vectors (in terms of cosine similarity, dot product, inverse Euclidean distance, or other geometric measures) for semantically similar input texts, even when those texts share few or no specific words. We obtained text embeddings for the episode and recalls using EmbeddingGemma (Vera et al., 2025), a pre-trained 308-million-parameter encoder-only transformer model that outputs coordinate vectors in a 768-dimensional embedding space.

To construct the trajectory for the *Atlanta* episode, we first converted our tabular annotations of the content present during each camera shot to a plain-text format. Separately for each shot, we concatenated the annotation text describing explicit on-screen occurrences and characters' implicit internal states. We then appended "Speech: " followed by the shot's dialogue, "Character speaking: " followed by the name of the speaker, and "Setting: " followed by the shot's setting. If any annotated fields were empty for a given shot (e.g., if no character was speaking at that time), we omitted them and their prefixes from the concatenated text. Next, we used EmbeddingGemma to transform the text snippet for each camera shot into a text embedding vector, resulting in a number-of-shots (478) by number-of-features (768) matrix. Since the camera shots varied in duration, we then resampled this series of embedding vectors to a consistent temporal resolution. We assigned each vector to the timepoint midway between the start and end of its corresponding camera shot (in seconds), and used linear interpolation to estimate a vector for each second of the episode. This yielded a number-of-seconds (1,455) by number-of-features (768) matrix—i.e., a 768-dimensional *trajectory*.

We then created similar trajectories for each participant's immediate and delayed recall of the episode. We began by splitting each recall transcript into a list of sentences, with the constraint that splits could not occur within reported speech (i.e., if a participant recounted a multi-sentence quote by a character in the episode, we treated the full quote as a single sentence-unit). We then parsed these sentences into overlapping sliding windows that spanned up to 5 consecutive sentences each. These windows grew to and shrank from their full size at the beginning and end of a participant's recall sequence—in other words, the first window contained only the first sentence of the participant's recall, the second contained the first two sentences, and so on. This ensured that all sentences from a participant's recall transcript appeared an equal number of times (5) in their sequence of sliding windows. Finally, we used EmbeddingGemma to transform each sliding window into a text embedding vector. Repeated for all recall transcripts, this yielded a separate trajectory characterizing each participant's recall from each of the two experiment sessions. The number of samples in participants' recall trajectories (i.e., the number of sliding windows parsed from their recall transcripts) ranged from 61 to 308 (mean: 136.68; standard deviation: 49.70).

Relating episode and recall embedding vectors

In prior work using our trajectory-analysis framework (e.g., Fitzpatrick et al., 2026; Heusser et al., 2021; Manning, 2021; Manning et al., 2022; Raccah et al., 2024), embeddings of the stimulus and participants' responses were constructed using a topic model (Latent Dirichlet Allocation; Blei et al., 2003) trained solely on the stimulus content. Here we elected to instead use a pre-trained transformer-based text embedding model due to the richer and more nuanced textual information these models are able to capture. (For example, these context-sensitive models can disambiguate the texts "Earn looked at Alfred" and "Alfred looked at Earn" in their representations, whereas bag-of-words models like LDA would represent the two identically.) However, these richer representations also introduce certain geometric nuisances we needed to account for in order to effectively relate the contents of participants' recalls to that of the episode they viewed. First, because modern transformer models are pre-trained on large and diverse text corpora, the representational capacities of their embedding spaces span an enormous breadth of semantic content.

Consequently, in preliminary analyses we found that the embeddings of our annotations and participants' recalls (which were all "about" the events, themes, and characters of a single *Atlanta* episode) were strongly anisotropic—i.e., they all occupied the same relatively narrow angular region of the model's overall embedding space. Additionally, because transformer models are sensitive to the syntactic structure of text, their embedding representations carry latent information about the text's linguistic register in addition to its semantic content. Accordingly, we found there was a consistent angular offset between embeddings of the episode annotations and those of participants' recalls, reflecting the fact that the annotations comprise grammatically well-structured written text while the recalls comprise transcripts of casual unrehearsed speech.

We corrected for these representational artifacts by mean-centering the episode and recall trajectories before performing analyses that involved measuring similarities between embedding vectors within or across trajectories. Conceptually, mean-centering a given set of embedding vectors (i.e., subtracting the across-vector average from each individual vector) serves to remove representations that are common across all vectors under consideration, and instead highlight representations that *vary* between them. For example, consider the sequence of embedding vectors comprising the trajectory of the *Atlanta* episode. Common across vectors within this sequence will be latent signals driven by how we chose to format the annotation text passed to the embedding model, the annotator's particular idiolect (i.e., idiosyncratic vocabulary and writing style), and any high-level overarching themes of the episode. *Not* common across these vectors are the finer-grained semantic representations that distinguish different events, scenes, and actions throughout the episode from each other. We found that removing these non-discriminative signals (to the extent they behave as additive offsets) via mean-centering helped reduce the embeddings' anisotropy (i.e., increased the range of measured between-vector similarities) and improved our ability to segment the episode trajectory into meaningful events. Similarly, mean-centering both the episode and recalls before relating their embedding representations helped mitigate broad-scale representational differences driven by their diverging registers, and enabled us to better relate them based on event-specific, content-driven patterns.

Our approach to mean-centering the recalls was guided by the particular latent representation we wanted to remove versus retain. In segmenting each participant's recall trajectory from each session into events (see *Event-segmentation*), our goal was to discriminate between embedding vectors strictly within a single recall sequence in order to identify temporally transient content-related patterns. We therefore centered each recall trajectory (in addition to the episode) on its own centroid to remove non-discriminative representations (e.g., register, broad-scale thematic content) and maximize the contrast between vectors within that single sequence. However, the analyses shown in Figures **REF TO DTW FIG** and **REF TO RDP FIG** entailed comparing sequences of recall events not only to the episode, but also to each other, both within- and across-participants, to examine whether within-participant similarities tended to exceed across-participant similarities. Here our goal was to remove representational artifacts that could work to bias our results towards significance. Specifically, participants' idiolects (i.e., particular speaking styles and sets of words with which they desecribe the episode) will influence the embedding representations of their recalls in ways that tend to be similar between their immediate and delayed recalls, but different across participants. To correct for this confound, we jointly centered each participant's set of immediate and delayed recall events on the two sequences' shared centroid (i.e., the average vector across all events from both sessions). This served to reduce the recall embeddings' anisotropy (by removing signals driven by high-level themes of the episode they recalled) and mitigate the effects of the spoken-recall register and participant-specific idiolect (both of which are shared between a participant's recalls from the two sessions) while preserving any genuine drift in the high-level content of each participant's recall over the delay (which per-session mean-centering would instead discard).

Event-segmentation

After constructing trajectories for the episode and each participant's immediate and delayed recall, we used hidden Markov models (HMMs; Baum and Petrie, 1966; Rabiner, 1989) to segment each trajectory into a sequence of discrete events. Given a number of states K, an HMM models an observed time series as a progression through K hidden states, each of which has a characteristic signature in the data (i.e., a pattern that timepoints belonging to that state tend to exhibit) and tends to persist across timepoints with occasional transitions rather than fluctuating freely. Because states persist, inferring the most probable state at each timepoint segments the observed time series into a sequence of events—i.e., spans of contiguous timepoints with the same most-probable state.

We segmented the episode trajectory into events using the HMM implemented in the BrainIAK toolbox (Kumar et al., 2021), which jointly learns a mean pattern for each of K states and the set of transitions between them via the Baum-Welch algorithm (Baum et al., 1970), with the added constraint that each state is visited exactly once in a strict left-to-right order (Baldassano et al., 2017). To select an appropriate number of states (i.e., events) for the episode, we used an optimization procedure we developed in prior work (Heusser et al., 2021) that aims to identify the single timescale of event representations emphasized most heavily in the episode's narrative structure. We began by segmenting the (mean-centered) episode trajectory into K events for each K between 2 and 50, inclusive. For each candidate segmentation, we then computed the cosine similarity between each pair of timepoints assigned to the same event, and between each pair of timepoints assigned to different events. When constructing these distributions of within- and across-event similarities, we restricted comparisons to pairs of timepoints separated by no more than the duration of the longest event in the candidate segmentation, so that the within- and across-event distributions were computed over comparable temporal distances and the across-event distribution was not dominated by distant timepoint pairs that we would expect to be dissimilar regardless of event structure. We then selected the candidate segmentation that maximized the Wasserstein distance between these within- and across-event similarity distributions. We found that the maximally distinctive and internally self-similar segmentation was yielded by $K = 20$ events. **(TODO: ADD SUPP FIG SHOWING WASSERSTEIN DIST AS FUNCTION OF K)** Finally, we averaged the embedding vectors over timepoints between each pair of adjacent event boundaries to obtain a sequence of stable event-average representations for the episode.

To segment participants' recall trajectories into sequences of events, we developed a novel HMM-driven procedure that grounded the recalls directly against the content of the episode participants viewed and recounted. Essentially, rather than re-learning separate sets of state representations from each participant's recalls (which tend to be shorter, noisier approximations of the episode content), we leveraged the state representations already learned from the episode to identify transitions between the episode events a given participant was (most probably) describing at a given point in time. First, we mean-centered the sequence of event-average embedding vectors for the episode, and each participant's sequence of per-window embedding vectors from each session, separately. For each given recall trajectory, we then computed the cosine similarity between the embedding vector for each recall window and the event-average embedding vector for each of the 20 episode events. This yielded a number-of-windows by number-of-episode-events (20) similarity matrix. We then treated the episode events as the hidden states of a hidden Markov model with fixed parameters, and treated each recall window's profile of similarities to the episode events, z-scored across all window-event pairs, as its distribution of emission scores over states. We imposed an ergodic transition structure (i.e., we allowed transitions from any state to any other state, since participants may recall the episode events out of order, and allowed states to be re-entered multiple times, since participants may revisit earlier events in their recalls) governed by separate additive transition penalties for forward-one transitions and all other transitions. Specifically, we imposed no penalty for remaining in the current state, a small penalty (1.5 normalized emissions units) for transitioning from a given state to the next (i.e., transitioning from recalling episode event n to recalling episode event $n + 1$), and a larger penalty (3.0 normalized emissions units) for all other transitions. Finally, we decoded the most probable sequence of episode-event labels using the Viterbi algorithm (Viterbi, 1967), collapsed runs of consecutive identically labeled recall windows into events, and averaged the embedding vectors over windows within each event. This procedure yielded, for each participant's recall from each session, a sequence of event-average embedding vectors and a set of labels denoting which episode event each recall event corresponded to.

2D path embedding

Data availability

All of the data analyzed in this manuscript may be found at https://github.com/ContextLab/memory-dynamics.

Code availability

All of the code for running our experiment and carrying out the analyses may be found at https://github.com/ContextLab/memory-dynamics.

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Author contributions

Competing interests

The authors declare no competing interests.